Sai Satwik Bikumandla

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PROFESSIONAL SUMMARY

Software Engineer with full-stack knowledge who builds production grade software applications with AI/ML capabilities.

EDUCATION

University at Albany, State University of New York — Aug 2024 - May 2026
Master of Science, *Computer Science*

Sri Indu College of Engineering & Technology (JNTUH) — Aug 2020 - Apr 2024
Bachelor of Technology, *Computer Science & Information Technology*

SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C++
Frameworks & Libraries: React.js, Next.js, Node.js, Flask, TensorFlow/Keras, YOLO, LSTM/RNN
Databases & Cloud: MongoDB, MySQL, Google Cloud SQL, Google Cloud Platform, AWS
DevOps & System Architecture: Docker, Kubernetes, CI/CD Pipelines, Microservices, System Design, API Design, Database Optimization
Web Design & Frontend: UI/UX Design, Responsive Design, CSS Animations, Figma, Wireframing, Prototyping, Design Systems, Accessibility (WCAG), HTML/CSS
Tools & Best Practices: Git, REST APIs, NextAuth, JWT, LaTeX, MATLAB, Gemini API, Code Review, Unit Testing, Agile/Scrum, Technical Documentation
Soft Skills: Knowledge Transfer, Communication, Scalability Optimization
Spoken Languages: Telugu (Native), Hindi (Fluent), English (Fluent), German (Actively Learning)

PROJECTS

LaTeX Resume Builder - Version-Controlled Resume Platform

- Designed and shipped a full-stack SaaS-style web application (Next.js, MongoDB, NextAuth) giving job seekers a GitHub-inspired resume workflow: multiple named repositories, commit snapshots, full version history, and rollback to any prior state - eliminating the problem of scattered, inconsistent resume files.
- Engineered a dual-render pipeline: an HTML live preview for zero-latency editing feedback alongside a server-side LaTeX compilation endpoint that exports polished, print-ready PDFs from any historical commit.
- Modeled resume versions as append-only, immutable snapshots (User → ResumeRepository → ResumeVersion), enforcing audit-trail integrity while keeping the UX approachable for non-technical users across 4 purpose-built pages.

Movie Recommendation System - Content-Based NLP Engine

- Built a content-based recommender over a dataset of 5,000+ films using TF-IDF vectorization and cosine similarity; applied a full NLP preprocessing pipeline (tokenization, lemmatization, stop-word removal) to improve feature signal quality.
- Evaluated recommendation relevance using Precision@K across K = 5, 10, and 20, enabling data-driven tuning of the similarity threshold and surfacing measurable lift over a random baseline.

Tennis Ball Detection & Tracking System

- Integrated Roboflow's pre-trained tennis ball detection model via REST API and built two production-quality tracking systems on top: a Basic Model with anti-jump filtering, exponential smoothing (a=0.85), and multi-frame velocity extrapolation; and an Advanced Model adding court-side A/B classification, perspective-aware overlays, and configurable color schemes per court surface type.
- Implemented SORT (Simple Online and Realtime Tracking) for multi-object tracking with custom bounding-box conversion, per-track trajectory trails (50-frame memory), and frame-by-frame annotation overlays displaying Ball ID, detection count, and active track count.
- Built a TrackingAnalytics engine that generates structured JSON output per session - capturing velocity peaks, total trajectory distance, side-time distribution, detection success rate, and position maps - exported alongside processed video.
- Deployed as an interactive Gradio UI with model selection (Basic vs. Advanced), configurable start/end time segments, court color scheme selection, live analytics summary, session run history, and ZIP download of processed video + analytics JSON.

Next-Word Predictor - LSTM Language Model

- Trained an LSTM language model (Embedding → LSTM 150 units → Softmax Dense) in TensorFlow/Keras on 2,000 IMDB reviews; built the full data pipeline - tokenization, sliding-window sequencing (length 20), padding, and one-hot encoding - over a 20,000-token corpus with an 80/20 train-validation split.
- Implemented checkpoint-based model persistence: training runs once and subsequent sessions load saved weights instantly, enabling iterative experimentation without redundant computation.

Web Prototype Generator - AI-Assisted Code Generation Tool

- Led the Gemini API integration layer, building the prompt engineering pipeline that converts natural-language client briefs into functional, responsive HTML/CSS/JS prototypes with auto-embedded Unsplash imagery; backend service layer built with Flask.
- Introduced in-source AI-attribution tagging and documentation standards adopted across the team codebase, ensuring transparency on human vs. AI-generated contributions.

WORK EXPERIENCE

Rethink UX | Website Design Intern — Oct 2022 - Dec 2022

- Designed and built client-facing website features in HTML, CSS, and JavaScript, taking assigned modules from requirements through delivery.
- Implemented access-control logic to restrict page features by user role.
- Wrote technical documentation for the features I built, making handoff easier for the team after the internship ended.
- Participated in code reviews with senior engineers and incorporated feedback to meet production standards.

StudyExperts | Programming Content Intern — Jun 2022 - Aug 2022

- Created programming tutorials and technical knowledge-base articles covering Python, Java & data structures, serving as self-service resources for the platform's users.
- Wrote, tested, and debugged code examples across multiple languages to ensure published content was accurate and runnable.
- Demonstrated that strong written communication and deep product understanding are force multipliers for small engineering teams - a lesson that now directly informs how I document and advocate for my own projects.

CERTIFICATIONS & ACHIEVEMENTS

- Google AI Essentials - Google — July 2026
- Building with the Claude API - Anthropic March 2026